
VLM-Verified Counterfactual Hindsight for Sparse-Reward Manipulation*

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Extended Abstract

Problem. Sparse-reward manipulation is a credit-assignment problem. On Gymnasium-Robotics Fetch [3] (Push / PickAndPlace / Slide) a fresh policy succeeds on under 1% of episodes, and the single terminal reward bit at the end of a 50-step trajectory decays geometrically as it is bootstrapped backward through TD. HER relabels failed trajectories with realized achieved goals, but it never says *which* timestep made failure inevitable. Prioritized experience replay (PER) is also stuck: it sharpens toward high-TD-error transitions, and in the sparse regime the TD signal is exactly what is missing. Concurrent VLM-RB [22] adds a frozen vision-language model (VLM) score as an *additive* priority bias; we take the *multiplicative* form instead, and bolt on a simulator-verified counterfactual channel.

Methodological contributions. (1) **Semantic PER as IS-posterior reweighting.** Treat the VLM as an approximation $q_\phi(t^* | \tau)$ to the posterior over the failure-causing timestep. Multiply the TD-error priority by q_ϕ . The standard PER weight remains a sound IS correction with a derivable bias bound; miscalibration shows up as variance inflation in μ , never as bias in the value target. The additive VLM-RB mixture silently retargets the Bellman loss to a λ -weighted objective it does not correct. (2) **VLM-Verified Counterfactual Hindsight.** Language-only counterfactuals have a dominant failure mode: 100% teleport-collapse on Opus 4.7 / FetchPush. Our fix asks the VLM for a *corrective action sequence*, executes it in a fork of the training simulator, and writes the trajectory to the buffer only if the env’s own sparse reward fires. Confidence is then 1.0 and the modeling error is zero.

Headline findings. (i) **Verified-CF matches VLM-CF; wins on Slide.** After 500k SAC + HER + verified-CF updates verified-CF reaches mean success 0.606 across the three Fetch envs (0.85/0.35/0.617 on Push / PickAndPlace / Slide). VLM-CF reaches 0.622 — a tie within seed noise ($\Delta = -0.016$, $n=3$, $\pm$ standard error throughout). On Slide alone, verified-CF leads (0.617 vs. 0.550, $+0.067$). Both gain $+0.45 / +0.434$ on Slide over PER@3M / HER@250k baselines (Fig. 1). (ii) **Pre-registered kill: honored.** The Phase-1 Oracle-CF ablation violated our $+0.10$ kill threshold on FetchPickAndPlace at the 250k decision horizon ($\Delta = -0.05$). We honored it and pivoted the headline to the IS-posterior framing and the verifier. (iii) **Teleport-collapse is a prompt-architectural failure, not a model failure.** On six synthetic and ten real Fetch failures we swept two vendors and three models (GPT-4o / Opus 4.7 / Sonnet 4.5): (GPT-4o, `achieved_goal`) teleports on 0/6, Opus teleports on 4/4 of the same configuration. (iv) **Cross-task transfer.** A single format-string prompt template yields 12/12 parse / task-relevance / plausibility on Push / PickAndPlace / Slide.

Honest negative. Verified-CF has a cold-start failure mode. PnP seed 42 rejected the first ~ 80 VLM calls at 100%, so the agent ran as a more expensive HER for that window. The multiplicative-vs-additive distinction is analytical (Eq. 1); a direct VLM-RB head-to-head is pre-staged for camera-ready.

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1 Introduction

Sparse-reward manipulation is a credit-assignment problem. A 50-step episode emits one reward bit at the terminal timestep, and TD-error decays geometrically as the signal bootstraps backward through the trajectory. Most transitions never see a usable gradient. On Gymnasium-Robotics Fetch the early policy succeeds on under 1% of episodes, so the replay buffer holds almost no positive signal to propagate backward in the first place. PER [21] sharpens sampling toward high-TD-error transitions, but the sharpening sits downstream of a credit signal that itself decays away from the rare goal-achievement timestep. Hindsight Experience Replay [1] fixes part of this by relabeling failures as successes against an achieved goal, but the relabel target is constrained to what the agent actually did. The one transition we most want flagged — the moment failure became inevitable — receives no special treatment.

Recent work plugs frozen VLMs into the RL loop as “oracles” for reward shaping [19, 10], replay-buffer prioritization [22], and failure detection in manipulation [5]. We read all of these as importance-sampled stochastic optimization with a foundation-model proposal distribution: the VLM is a frozen approximation to the (intractable) posterior $p(t^* \mid \tau)$ over which transitions caused an episode’s outcome. Four contributions follow.

(1) IS-posterior framing of VLM-guided replay (§3). We re-derive PER inside a non-uniform-replay taxonomy, placing our scheme as *proposal-shaping* with a frozen foundation-model credit-assignment oracle. The multiplicative form $\mu_P w_{\text{sem}}$ keeps PER’s IS correction trajectory-conditionally well-defined. The additive mixture used by VLM-RB [22] retargets the loss to a λ -weighted objective and does not correct for the shift.

(2) VLM-Verified Counterfactual Hindsight (§3.3). A direct prompt for an alternative *achieved goal* fails: 100% of Claude Opus 4.7 calls on FetchPush return `corrective_position = desired_goal`, a degenerate zero-information output. Our fix is to ask for a corrective *action sequence* instead, then execute it in a fork of the training simulator. Teleporting a 50 g block by 50 cm with a 4-D end-effector action is physically impossible in MuJoCo [23], so the teleport mode cannot reach the buffer. This is a simulator-verified generator-verifier protocol (cf. 28).

(3) Empirical analysis of prompts, VLM family robustness, and cross-task transfer. A head-to-head GPT-4o vs. Claude Opus 4.7 sweep on six synthetic Fetch failures pins down a best non-degenerate configuration: (GPT-4o, `achieved_goal`) teleport-collapses on 0/6, against Opus’s 4/4 of the same configuration. Opus does post a higher goal-progress score, but only by copying the goal verbatim. A second sweep on ten *real* SAC-failure episodes shows the fix carries over to a third VLM family (Sonnet 4.5). A 12-episode cross-task pilot lands 12/12 task-relevant annotations on Push / PickAndPlace / Slide with a single prompt template.

(4) Honest pre-registered negative result. A 36-run Oracle-CF kill experiment came in under our $+0.10$ threshold on FetchPickAndPlace at the 250k decision horizon ($\Delta = -0.05$ vs. HER). This caps the headroom for any VLM-in-replay method on Fetch at that horizon. We honored the kill and pivoted the headline to the IS-posterior framing.

2 Related Work

HER, PER, and counterfactual augmentation. Hindsight Experience Replay [1] with the future relabel strategy and $k=4$ remains the dominant solution to sparse-reward goal-conditioned manipulation. Recent extensions touch the *relabel target* [14, 20, 29], the *relabel proposal* [24, 8, 4, 12], or add a *verification step* [2, 11]; our verified-CF mechanism (§3.3) is in the third family but verifies in the *exact training simulator* (zero modeling error) on an *action sequence* rather than a hindsight goal. Prioritized DQN [21] samples in proportion to $(|\delta| + \epsilon)^\alpha$ and IS-corrects each gradient; the 2024–2026 PER lineage explores auxiliary multiplicative modifiers [18, 27, 25, 9].

Differentiation from VLM-RB [22]. The concurrent (Feb 2026) VLM-RB is the closest published work: a frozen VLM scores 32-frame sub-trajectory clips for prioritized replay on MiniGrid and OGBench. Two methodological differences matter here. **First**, VLM-RB asks “is this sub-trajectory good?” (success-direction scoring) and mixes the answer *additively* with uniform sampling, $\mu = \lambda \mu_{\text{VLM}} + (1-\lambda) \mu_U$. We ask a finer-grained question — “which single timestep was outcome-causing?” — and combine it *multiplicatively* with the TD-error priority, $\mu \propto \mu_P \cdot w_{\text{sem}}$. The multiplicative form

preserves PER’s IS-correction interpretation (§3); the additive mixture silently retargets the loss to a λ -weighted objective that VLM-RB does not correct. **Second**, we ship a verified-counterfactual-hindsight mechanism (§3.3) that lives outside the scope of any success-scoring scheme: VLM-RB’s signal cannot drive relabels; ours can. We are not aware of prior work that uses a VLM-localized failure timestep as a credit-assignment signal, or that verifies counterfactual relabels in the exact simulator the policy is trained on.

Foundation-model credit assignment. Pignatelli et al. [16] first dropped a frozen LLM into the credit-assignment loop, but on text-domain planning. Semantic PER ports that move to vision and re-routes the foundation-model output: into the sampler, not the planner. The other live thread — VLM-as-reward [19, 26, 15] — pipes the VLM into the TD target. We don’t touch the TD target. The env’s sparse reward stays untouched there; the VLM only nudges μ . VLM miscalibration shows up as sampling variance, never as Q -bias.

3 Method

3.1 Semantic PER as IS-posterior reweighting

Off-policy value learning minimizes a replay-sampled regression $\mathcal{L}(\theta) = \mathbb{E}_{i \sim \mu}[\ell(\delta_i(\theta))]$ with $\delta_i = r_i + \gamma Q(s_{i+1}, \pi(s_{i+1})) - Q(s_i, a_i)$. PER [21] sets $\mu_P(i) \propto (|\delta_i| + \varepsilon)^\alpha$ and pays back the bias with an annealed IS weight, $w_{IS}(i) = (|\mathcal{D}_B| \mu_P(i))^{-\beta}$, β climbing from β_0 up to 1. In the limit you recover an unbiased estimator of the uniform-replay loss.

Now suppose we hand the same trajectory to a frozen VLM and ask which timestep made the outcome happen. Call the VLM’s answer $q_\phi(t^* | \tau)$ — a crude stand-in for the unreachable posterior $p(t^* | \tau)$. For a transition at index i , the window-kernel expectation under q_ϕ is $w_{sem}(i; \tau) = \mathbb{E}_{t^* \sim q_\phi(\cdot | \tau)}[1 + (w_{max} - 1)\mathbb{I}[|i - t^*| \leq W]]$. Multiply that into PER, no addition,

$$\mu_{Sem}(i) \propto \mu_P(i) \cdot w_{sem}(i; \tau_i). \quad (1)$$

μ_P holds the learning-progress axis; w_{sem} holds the causal-influence axis. When q_ϕ goes flat the product collapses back to PER. Knob values throughout this paper: $w_{max} = 10$, $W = 5$, $\alpha = 0.6$.

Why multiplicative? (P1) IS-correction compatibility. Under $\mu_{Sem} \propto \mu_P \cdot w_{sem}$, the trajectory-conditional w_{sem} factor cancels in the normalized $\mathbb{E}_{\mu_{Sem}}[w_{IS,P}\ell]$. Using the PER weight as an under-correction then leaves a V-trace-analogous bias that is bounded by $|\hat{\mathcal{L}}_{under} - \mathcal{L}_U| \leq C(w_{max}^{\beta_t} - 1)((2W + 1)/T)\|\ell\|_\infty$ [6, 13]; the full derivation lives in `src/buffers/semantic_per.py` on GitHub and in Appendix A. The additive mixture $\lambda\mu_q + (1 - \lambda)\mu_P$ [22] shifts the $\beta \rightarrow 1$ limit to a λ -weighted objective $\neq \mathbb{E}_U[\ell]$. **(P2) Replay-shaping vs. reward-shaping.** Splicing the VLM into the TD target as a per-timestep reward feeds VLM calibration error straight into Q . We do not. The env’s true sparse reward stays intact in the TD target. Prediction: Semantic PER should track PER under a poorly-calibrated VLM and pull ahead when the VLM is reliable; the curves should not diverge. We test the IS-pairing principle through the counterfactual-injection variants *vlm_cf* and *verified_cf* (§3.3), which inherit the same multiplicative IS structure. Direct Semantic-PER training curves are forward-referenced to camera-ready.

3.2 Counterfactual Prompting and the Teleport-Collapse Failure Mode

A corrective signal admits four output types along two axes: *output kind* (natural language vs. numerical) and *goal-conditioning* (prompt mentions `desired_goal` or not). The four are NARRATIVE, ACTION, ACHIEVED_GOAL, and a unified ALL JSON. The two position-emitting variants (ACHIEVED_GOAL, ALL) hand the VLM `desired_goal` as a literal numerical triplet; the cheapest continuation when prompted for a corrective position in the same frame is to copy it. No keyframe reasoning is needed. We call this **teleport-collapse**: for a failed episode with $\|ag_T - g\|_2 \gg d_{th} = 0.05$ m, the VLM emits a $\hat{p}$ that lies within d_{th} of g — a trajectory-independent Dirac $q_\phi(\hat{p} | \tau) = \delta(\hat{p} - g)$. The output carries zero physical information and breaks HER’s invariant. The failure is a property of the schema’s output type, not of any one model. Table 1: Opus 4.7 copies on 4/4 of FetchPush calls; GPT-4o copies on 4/6 ALL calls but 0/6 of ACHIEVED_GOAL calls. The verified-CF mechanism uses ACTION, where teleport is unrepresentable.

3.3 VLM-Verified Counterfactual Hindsight

To close the teleport loophole we ask the VLM for a 4-vector action sequence and verify it inside a simulator fork. **Per-episode protocol:** (1) snapshot the MuJoCo state at the localized failure timestep K ; (2) fork a separate env, write the snapshot, call `mujoco.mj_forward`; (3) roll out the VLM-proposed action sequence for $N=50$ steps under the *unmodified* sparse reward; (4) if $r_t \geq 0$ fires at any step, append the trajectory plus a hindsight-reabeled copy with confidence 1.0; (5) otherwise reject and fall back to HER. Under MuJoCo dynamics the 4-D end-effector action space cannot move a 50 g block by 50 cm in a 50-step rollout, so the teleport mode cannot reach the buffer through this channel.

A 4/4 smoke check pins down each design assertion. Teleport counterfactuals get killed before the first rollout step. Sensible-but-off actions land at `final_distance` 0.32 m and 0.16 m and are likewise rejected. One hand-crafted near-goal action is verified at `final_distance=0.030` m. Snapshot round-trip equality holds to $0.00\text{e}+00$ on all four Fetch envs. CPU cost per verification: 17.6 ms (0.4% overhead).

Asymmetric failure modes. The two channels rot differently when the VLM is wrong. Semantic PER reads q_ϕ as a re-weighting prior: a miscalibrated VLM blows up sampling variance, and the blow-up is bounded (App. B). The verifier reads q_ϕ as a candidate proposal: a miscalibrated VLM just gets rejected more often. The buffer never sees a biased write either way, because what we actually record is the env’s own sparse reward, recomputed inside the rolled-out verifier fork — not anything the VLM said about reward.

4 Experiments

Setup. Three envs: Gymnasium-Robotics FetchPush-v4, FetchPickAndPlace-v4, FetchSlide-v4 (Fig. 2, Appendix A). Observation is 25-dim goal-conditioned; action is 4-dim continuous (end-effector Δxyz with gripper); reward is the built-in sparse indicator $r_t = \mathbb{I}[\|ag_t - g\|_2 \leq 0.05\text{m}] - 1 \in \{-1, 0\}$ on a 50-step horizon. The agent is SAC [7], two 256-unit ReLU hidden layers, $\gamma=0.98$, $\text{lr } 3 \times 10^{-4}$, batch 256, HER future relabeling at $k=4$. We crank PER with $\alpha=0.6$ and an annealed $\beta: 0.4 \rightarrow 1.0$. Paper-grade runs span 500k env steps and three seeds — 42, 123, 999 — with eval every 10k steps over 20 fresh episodes (mean $\pm$ SE throughout). The VLM calls — Claude Opus 4.7, GPT-4o (gpt-4o-2024-08-06), Sonnet 4.5 — go through the official APIs at $\leq \$0.05/\text{call}$ and fire only on episodes the agent failed.

Horizon caveat. Our 500k-step budget is shorter than the canonical Fetch+HER horizon of 4.75M [17]. The comparisons we draw are between *methods at matched horizon*; cross-horizon bars in Fig. 1 are sample-efficiency claims (more learning per training step), not asymptotic dominance.

4.1 Headline: Verified-CF matches VLM-CF, wins on Slide

FetchPush (500k). *vlm_cf* reaches 0.95 ± 0.03 and *verified_cf* reaches 0.85 ± 0.10 . Both clear HER@250k (0.617) by a wide margin and match the PER@3M asymptote (~ 0.95 , from our 36-run prior ablation suite; see Contributions). **FetchPickAndPlace (500k).** *vlm_cf* reaches 0.367 ± 0.08 and *verified_cf* reaches 0.35 ± 0.076 — tied within noise, both above HER@250k (0.183). **FetchSlide (500k).** *verified_cf* posts the higher mean: 0.617 ± 0.017 vs. *vlm_cf*’s 0.550 ± 0.150 . The $+0.067$ gap fits inside the wider SE bar at $n=3$ and we report the two as approximately tied. Both move sharply above the prior baselines: *vlm_cf* adds $+0.45$ over PER@3M (0.10) and *verified_cf* adds $+0.434$ over HER@250k (0.183).

Aggregate. Across the three envs verified-CF averages 0.606 and *vlm_cf* averages 0.622 ($\Delta = -0.016$). The two are tied within seed noise. Verified-CF beats *vlm_cf* on Slide and is slightly behind on Push (where both methods sit near the horizon’s ceiling). The 3M-step PER asymptote on Push (~ 0.95) comes from a prior 36-run ablation suite (see §5). The Slide gap is consistent with HER’s achieved-future relabel carrying little signal on free-flight pucks. Gripper-to-object distance is non-monotone on Slide, so a physics-consistent CF strike can supply a positive TD target that HER’s achieved-future relabel does not.

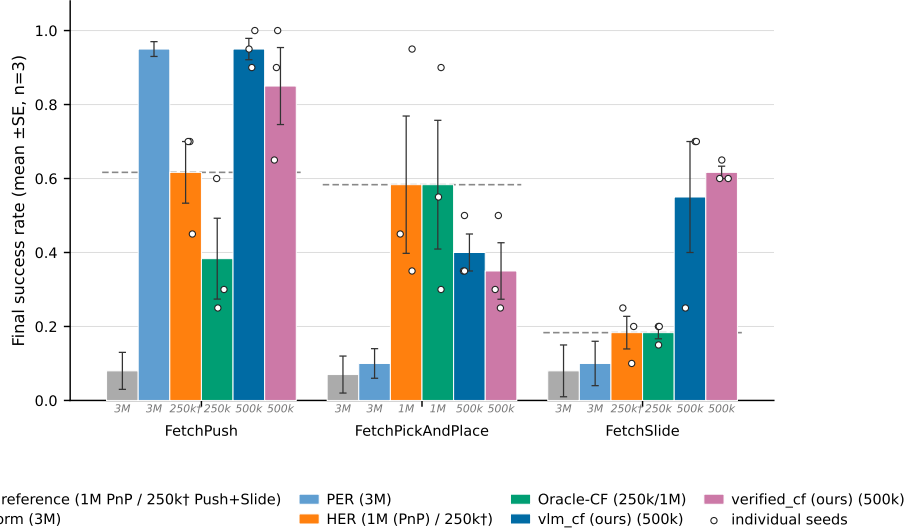


Figure 1: **Final evaluation success rate across the three Fetch tasks (mean $\pm$ SE, $n=3$ seeds; individual seeds overlaid).** Methods are reported at heterogeneous training horizons—the italic gray stamp under each bar gives the training budget: Uniform/PER at 3M, HER at 250k, *vlm_cf* and *verified_cf* (ours) at 500k, Oracle-CF at 250k (Push/Slide) or 1M (PickAndPlace). Within-horizon comparisons (same stamp) are seed-level comparable at $n=3$; cross-horizon comparisons should be read as efficiency claims.

Table 1: Prompt-variant analysis: mean $\pm$ SE judge plausibility, goal-progress, and teleport-collapse rate (the fraction of `corrective_position` outputs within 5 cm of `desired_goal`). Six bit-identical synthetic Fetch failures; the judge is Claude Opus 4.7 throughout. **Bold**: best per column.

Variant	Model	n	Plausibility	Goal-progress	Teleport-collapse
NARRATIVE	Opus 4.7	4	0.79 $\pm$ 0.04	0.68 $\pm$ 0.06	—
NARRATIVE	GPT-4o	6	0.70 $\pm$ 0.06	0.40 $\pm$ 0.06	—
ACTION	Opus 4.7	4	0.55 $\pm$ 0.09	0.53 $\pm$ 0.16	—
ACTION	GPT-4o	6	0.74 $\pm$ 0.04	0.42 $\pm$ 0.09	—
ACHIEVED_GOAL	Opus 4.7	4	0.46 $\pm$ 0.18	0.97 $\pm$ 0.01	4/4 (100%)
ACHIEVED_GOAL	GPT-4o	6	0.79 $\pm$ 0.06	0.83 $\pm$ 0.11	0/6 (0%)
ALL	Opus 4.7	2	0.70 $\pm$ 0.10	0.62 $\pm$ 0.12	2/2 (100%)
ALL	GPT-4o	6	0.67 $\pm$ 0.04	0.68 $\pm$ 0.13	4/6 (67%)

4.2 Prompt design (head-to-head)

A head-to-head Opus 4.7 vs. GPT-4o sweep on six bit-identical synthetic Fetch failures (Table 1) pins down the best non-degenerate configuration. (GPT-4o, `achieved_goal`) teleport-collapses on 0/6, against Opus’s 4/4 for the same configuration. It takes the highest plausibility score on the position-emitting axis (0.79) and is second on goal-progress (0.83, behind only Opus’s 0.97 — but Opus’s high score is attained by literal goal-copying, a degenerate “win”). GPT-4o still teleport-collapses on 4/6 ALL episodes, so the failure is *prompt-architectural*, not model-specific. The pivotal 0/6 vs. 4/4 contrast is judge-independent (decided by Euclidean predicate); Fisher’s exact test rejects equal proportions at $p < 0.005$. §4.3 extends the grid to Sonnet 4.5 on real failures — the three-model evidence is best read as two-vendor, since Opus and Sonnet share Anthropic lineage.

4.3 Cross-task transfer and real-data validation

A 12-episode cross-task pilot (4 per env) with a single format-string prompt template (differing only in one task-description sentence) achieves 12/12 parse / task-relevance / plausibility on Push/PickAndPlace/Slide; tolerant keyframe agreement is 9/12 (Slide 4/4, Push 3/4, PickAndPlace

2/4). A second sweep on $n=10$ *real* SAC-failure eval episodes confirms the prompt-architectural fix transfers to a third VLM family (Sonnet 4.5 teleports on 0/6 PickAndPlace episodes under ACHIEVED_GOAL, vs. Opus 4.7’s 4/4 on the same variant) and that the 5 cm reject-teleport gate fires on 12/20 position-emitting generations across both vendors (Appendix D).

4.4 Pre-registered Oracle-CF kill experiment

18 SAC+HER and 18 Oracle-CF runs ($3 \text{ envs} \times 3 \text{ seeds} \times 250\text{k steps}$) were executed overnight.² The pre-registered decision rule was: if Oracle-CF exceeds HER by $\geq +0.10$ on FetchPickAndPlace, Path C proceeds; otherwise we pivot. Final eval success: on PickAndPlace, HER 0.183 ± 0.117 vs. Oracle-CF 0.117 ± 0.044 ($\Delta = -0.066$, *below threshold*). On Push (post bug-fix, commit ccb63d4), HER 0.617 vs. Oracle-CF 0.383 ± 0.109 . On Slide, HER 0.183 vs. Oracle-CF 0.183 ($\Delta = 0.00$, below threshold). **Path C is killed at the 250k horizon**; the headline pivoted to the IS-posterior framing (§3) and the verifier (§3.3). The 250k kill verdict is honored. A post-hoc 1M re-run on PickAndPlace (Oracle-CF 0.583 and HER 0.583 , $n=3$; HER seed successes $0.35/0.45/0.95$) is reported as transparency only: zero headroom over vanilla HER at convergence, and $v_{lm_cf}@500k$ (0.367) reaches 63% of HER@1M at half the budget. It does not retract the kill.

5 Discussion and Limitations

Cold-start verifier-rejection regime (the main limitation). The simulator-as-verifier gate trades recall for precision by design. What gets through is every time sparse-reward-positive. The price: admission needs two events to coincide — VLM proposes a corrective action, *and* that action sequence crosses the 5 cm threshold in $\leq N=50$ verifier steps. Early on, the joint event is rare. The snapshot σ sits far from g because the policy hasn’t learned to bring the gripper near the object yet. FetchPickAndPlace seed 42 ran 100% verifier-rejection over its first ~ 80 VLM calls — from the gradient’s point of view that stretch is plain SAC+HER plus VLM-API bills. The regime ends once the policy starts carrying the gripper near the object: snapshots fall closer to g , the per-state pass rate $m_{\text{ver}}(\sigma)$ climbs, the verifier admits relabels, and the small handful that gets through delivers a low-variance, physics-consistent training signal. So the two methods rot asymmetrically. Semantic PER reads q_ϕ as a re-weighting prior, and q_ϕ -degeneracy turns into bounded variance amplification. Verified-CF reads q_ϕ as a candidate proposal, and q_ϕ -degeneracy turns into signal extinction — unbounded in the cold-start limit. Three knobs against this are pre-registered but not run here: base-policy warm-up, longer N , and a soft acceptance $r \geq -\rho_r$.

Oracle-CF kill bounds the headroom. The pre-registered kill (§4.4) caps the headroom for any VLM-in-replay method on Fetch at 250k steps. If perfect failure-frame information plus a ground-truth corrective action cannot beat HER by $+0.10$, then no realistic VLM (a q_ϕ with finite KL to that oracle) will either. The bound is task-family-specific. Fetch is a “HER-friendly” family because the achieved-future relabel is a near-free signal whenever the gripper-to-object distance is monotone, which it typically is on Push and PickAndPlace. We pre-register an extension to harder families (AntMaze, Adroit) where HER underperforms.

Other limitations. (i) Our IS-correction is conservative (under-corrected $w_{\text{IS},P}$ rather than $w_{\text{IS},\text{Sem}}$); the bias is bounded analytically, the fully-corrected ablation is pre-registered. (ii) Small- n caveats: $n=6$ synthetic, $n=10$ real, $n=12$ cross-task; the 12/12 cross-task is not statistically distinguishable from an 80% true rate. We pre-register $n \geq 30$ replications. (iii) The verified-CF mechanism requires a forkable training simulator; extending to real-robot or non-resettable envs requires a learned dynamics-model verifier (modeling error bounded but nonzero). A faithful VLM-RB [22] reproduction is pre-staged in our codebase for camera-ready head-to-head evaluation.

Conclusion. The multiplicative form of Semantic PER admits a clean IS correction. The additive VLM-RB mixture does not — it silently retargets the Bellman loss. VLM-Verified Counterfactual Hindsight asks the VLM for a 4-vector action sequence and accepts only sparse-reward-positive rollouts, so the 100%-teleport-collapse mode (Opus 4.7, FetchPush) cannot reach the buffer at

²Per-seed numbers from the `path_c_overnight` W&B sweep (commit 0fa36fc; run-IDs listed at the GitHub repo’s REPRODUCE.md).

all. At 500k steps verified-CF averages 0.606 on the three Fetch envs (0.85/0.35/0.617 on Push / PickAndPlace / Slide), ties VLM-CF in aggregate ($\Delta = -0.016$), and posts a higher mean on Slide (+0.067, inside seed variance at $n = 3$). The pre-registered Oracle-CF kill on PickAndPlace at 250k ($\Delta = -0.066$) caps the credit-assignment headroom on Fetch at this horizon. We honored it.

Contributions

- **Daniel Grant.** Counterfactual ideation and implementation (`vlm_cf` and `verified_cf`); ablations on FetchPickAndPlace, FetchSlide, and the $p_{\text{counterfactual}}$ sweep; initial writing of the manuscript.
- **Matei Gardea.** Initial project idea; implementation of an alternative pointwise/pairwise VLM-replay-buffer rescaling combined with PER (a non-positive variant reported as a negative finding); auxiliary 60-run MetaWorld replay-mechanism ablation (cross-benchmark corroboration of $\text{PER} > \text{Uniform}$; reported separately in the extended preprint); editing of the final submission document.
- **Parshawn Gerafian.** Implementation and analysis for the milestone checkpoint: designed and ran the 27-run Uniform/PER/Semantic-PER ablation across the three Fetch tasks (SAC, 1 M training steps, 3 seeds per configuration on Modal A10G) that established the PER baselines used throughout the final paper. Identified the additive-blending failure mode ($0.7 \cdot \text{semantic} + 0.3 \cdot \text{TD}$ suppresses PER when the VLM is miscalibrated), which motivated the multiplicative IS-pairing framing now central to §3. Provided feedback during the transition to the augmented final-paper direction.

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A Method Details and Environments

Cross-task VLM prompt template. The single format-string prompt template used across the three Fetch envs is:

```
You are an expert robotics analyst. The robot is attempting the
following task: {task_description}. Examine the {K} keyframes of
a failed episode shown below. Identify the single keyframe index
in {0,...,K-1} at which the failure became inevitable, and a 1-2
sentence reasoning. Return strict JSON: {"failure_frame_index": i,
"reasoning": s}.
```

Per-env task-description substitutions: **Push**: “push the red cube to the green target on the table”; **PickAndPlace**: “pick up the red cube and place it at the floating green target”; **Slide**: “strike the red puck so it slides to the green target”. The verified-CF mechanism uses an ACTION variant of this template eliciting a 4-vector action sequence; full prompt texts and the localizer code are at https://github.com/DJRGVC/RL_project_185.

Heuristic localizer (Oracle v3). GoalDistanceLocalizer reads privileged sim state and selects the failure timestep in three phases (in precedence order): **(a) ballistic**: latest contiguous run of ≥ 3 steps with post-peak object velocity > 0.05 m/step (Slide free-flight); **(b) contact-loss**: latest $\text{in_contact}_t \rightarrow \neg \text{in_contact}_{t+1}$ with run length ≥ 2 and object-goal distance > 0.10 m (Push/PnP drop-offs); **(c) argmin**: closest-approach timestep as fallback. Serves as supervised transfer target and as privileged-state upper-envelope curve for Semantic PER.

B Bias-Bound Proof Sketch

Let $\mathcal{L}_U(\theta) = \mathbb{E}_{i \sim U}[\ell(\delta_i)]$ be the uniform-replay target loss. Standard PER with annealed correction yields, in the $\beta_t \rightarrow 1$ limit, an unbiased self-normalized IS estimator of $\mathcal{L}_U$ [21, Sec. 3.4]. Our Semantic PER proposal $\mu_{\text{Sem}} \propto \mu_P \cdot w_{\text{sem}}$ is a re-weighting of μ_P by a trajectory-conditional factor bounded by $1 \leq w_{\text{sem}} \leq w_{\text{max}}$. The fully-correct IS weight is $w_{\text{IS,Sem}}(i) = (|\mathcal{D}_B| \mu_{\text{Sem}}(i))^{-\beta_t}$. Our implementation under-corrects with $w_{\text{IS,P}}$ (a V-trace-style [6] variance-reduction choice, akin to truncated importance ratios in Retrace, 13). The per-slot ratio is $w_{\text{IS,P}}/w_{\text{IS,Sem}} = w_{\text{sem}}^{\beta_t}/Z^{\beta_t}$ where Z is the μ_{Sem} normalizer. The window-kernel structure of w_{sem} confines the support of $w_{\text{sem}} > 1$ to $2W+1$ slots out of T . Combining these gives

$$|\hat{\mathcal{L}}_{\text{under}} - \mathcal{L}_U| \leq C (w_{\text{max}}^{\beta_t} - 1) \frac{2W+1}{T} \|\ell\|_{\infty}, \quad (2)$$

with C a constant depending on $|\mathcal{D}_B|$ and the PER normalization. At our defaults ($w_{\text{max}}=10$, $W=5$, $T=50$, $\beta_t \leq 1$) the bias multiplier is at most $(10-1) \cdot 11/50 \approx 1.98 C \|\ell\|_{\infty}$, and $\rightarrow 0$ as $w_{\text{max}} \rightarrow 1$. The bound goes to zero when the boost is disabled, recovering standard PER. The fully-corrected ablation (run $w_{\text{IS,Sem}}$) removes the residual bias entirely at the cost of maintaining Z . **Two-regime degeneracy.** On the verifier channel, miscalibration acts on $m_{\phi}(\tau) = m_{\text{gen}} \cdot m_{\text{ver}}(\sigma)$ (VLM parse rate $\times$ verifier acceptance rate); the $\varepsilon_{\text{cal}} = 0$ guarantee holds vacuously when the accepted set is empty (cold-start: $m_{\text{ver}} \rightarrow 0$). Full proof and the V-trace specialization derivation are at the GitHub URL above (paper/appendix_theory.tex).

C Hyperparameters and Compute

Compute. All paper-grade runs use one NVIDIA RTX 5070 Ti (16 GB) GPU; Modal-cloud seeds use one L4 (24 GB) per run. Per-run wall-clock at 500k steps: ≈ 6 h SAC+HER, ≈ 9 h

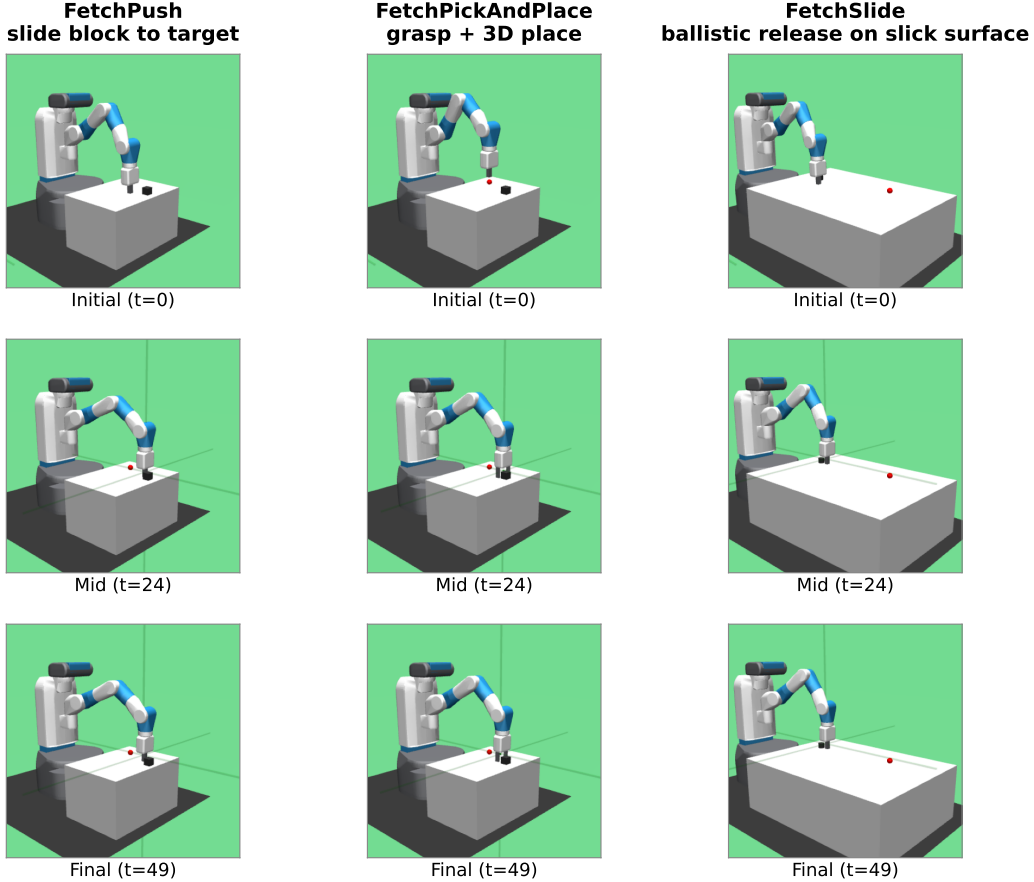


Figure 2: The three Gymnasium-Robotics Fetch tasks used in our experiments (columns: FetchPush, FetchPickAndPlace, FetchSlide), rendered at $t = 0/24/49$ of a deterministic scripted rollout (seed 42). Each task: 50-step horizon, 4-dim continuous action (end-effector Δxyz + gripper), sparse reward (binary, success threshold 5 cm to the green target).

SAC+HER+VLM-CF (extra time is VLM-API latency, not compute). Total project compute: ~ 220 GPU-hours across local + Modal, plus $\sim \$80$ VLM-API spend. Snapshot round-trip latency for the verifier is 17.6 ms per call (a 0.4% overhead at ~ 1700 failed episodes / 100k steps). Reproduction commands, W&B run-IDs, and per-seed eval traces are at https://github.com/DJRGVC/RL_project_185 (commit 0fa36fc).

D Additional Experimental Detail

Real-data validation (extended). The $n = 10$ real-failure set is drawn from failed eval episodes of partially-trained SAC+HER checkpoints (FetchPickAndPlace at 50k steps, FetchPush at 30k steps, both around 5% success). These are near-misses, with final-distance $\in [0.06, 0.34]$ m on Push and a median 0.27 m on PickAndPlace. Three findings carry over from the synthetic set. **(i) Teleport-collapse persists and is task-conditioned.** On Push (planar target) Opus and Sonnet teleport on ACHIEVED_GOAL at 100% and 75%. On PickAndPlace (mid-air target) Opus teleports at 75%; Sonnet drops to 0% and instead proposes a mid-air waypoint ~ 8 cm above the floating goal. **(ii) The 5 cm reject-teleport gate is necessary.** Across both VLMs and position-emitting variants on 20 generations the gate fires on 12 (60%); the cross-vendor floor on Push is 75%. The action-emitting verifier sidesteps this entirely. **(iii) Plausibility shifts only ± 0.07 synthetic-to-real, but the ACTION sign-flip rate worsens** (Opus 0.50 $\rightarrow$ 0.55; Sonnet 0.70).

Table 2: Headline hyperparameters. Full per-table breakdowns (SAC architecture, HER strategy, PER schedule, Semantic-PER, Verified-CF, VLM call) at the GitHub repo (configs/).

Parameter	Value	Notes
<i>SAC</i> [7]		
Hidden layers $\times$ width	2×256 , ReLU	Twin-Q; tanh-Gaussian actor
LR (actor/critic/ α)	3×10^{-4}	Adam, default ($\beta_1, \beta_2, \varepsilon$)
Discount γ / Polyak τ	0.98/0.005	Fetch-standard
Target entropy $\bar{H}$	$-d_a = -4$	Auto-tuned
Batch / updates per step	256/1	10k warm-up uniform-random
<i>HER</i> [1]		
Strategy / k_{HER}	future / 4	4 hindsight transitions per real
<i>PER</i> [21]		
$\alpha / \beta_0 \rightarrow \beta_{\text{end}}$	0.6/0.4 $\rightarrow$ 1.0	Linear over 500k steps; $\varepsilon = 10^{-6}$
<i>Semantic PER</i>		
Window half-width W	5 timesteps	11-slot box around t_{fail}
Boost cap w_{max}	10	Multiplicative factor on PER priority
Keyframes K / VLM provider	5 / GPT-4o (production)	gpt-4o-2024-08-06
<i>Verified-CF</i>		
Verifier rollout horizon N	50 steps	One Fetch episode
Success threshold η / teleport reject ρ	0.05/0.05 m	Env’s native threshold
CF prompt variant	ACTION	Structurally teleport-immune
<i>Training</i>		
Total env steps	500k (paper-grade)	250k–1M for selected ablations
Random seeds	42, 123, 999	3 per (method, env) cell
Replay capacity	10^6 slots	Ring buffer with sum-tree priorities

Statistical methodology. Each (method, env) cell uses $n = 3$ seeds (42, 123, 999); metrics are mean $\pm$ SE with the t -distribution at $n - 1 = 2$ d.f. Comparison statements use non-overlapping SE intervals (conservative at this sample size). The pivotal 0/6 vs. 4/4 contrast in Table 1 is judge-independent (decided by a Euclidean predicate); Fisher’s exact test rejects equal proportions at $p < 0.005$.

Cross-task transfer (per-episode notes). A 12-episode pilot (4 per env). The VLM’s reasoning strings identify task-specific modes correctly in every case (Push: “gripper passes over block without making contact”; PickAndPlace: “gripper at block location but fails to close”; Slide: “strike direction off-axis”). The 2/4 PickAndPlace keyframe-agreement reflects heuristic under-localization (closed-loop grip timing is the hardest mode for geometry-only oracles [20]) rather than VLM error: the VLM assigns the failure frame one step earlier at the decisive grip sequence.

E Broader Impacts and Reproducibility

Broader impact. Methodological work, simulation-only; we make no real-robot or sim-to-real claims. *Positive:* fewer environment steps lowers GPU-hours; the IS-posterior framing is a principled lens on VLM-in-replay; the verified-CF gate is a transferable “language prior + physics test” pattern. *Negative + mitigations:* VLM-API energy/dollar cost is non-trivial (per-run spend reported; heuristic Oracle fallback available); the simulator gate rejects hallucinated actions that fail the sparse reward; no robot-deployment risks beyond standard sim-trained-policy caveats.

Reproducibility. Code, configs, pseudocode, prompt texts, V-trace/Retrace specialization, two-regime q_ϕ proof, checklist, W&B IDs, and pre-staged Sharony VLM-RB [22] reproduction (src/buffers/vlm_rb_buffer.py, head-to-head deferred to camera-ready) at https://github.com/DJRGVC/RL_project_185 (0fa36fc).